

CHANDRASHEKAR BALA

Cybersecurity Researcher | Offensive Security | Security Engineering | AI Security

 chandrashekar-bala@protonmail.com

 +91 9133093202

 Hyderabad, Telangana

 in/chandrashekar-bala

 github.com/chandrashekar-bala

 chandrashekar-bala.github.io

Available for cybersecurity opportunities

PROFESSIONAL SUMMARY

Cybersecurity professional with hands-on independent experience in offensive security, vulnerability assessment, penetration testing, security research, and systems security. Experienced in reconnaissance, service enumeration, vulnerability validation, controlled exploitation, privilege escalation, attack-path analysis, web application security testing, exploit research, and proof-of-concept development using Linux-based security tooling. Strong programming and systems background in C, Python, Bash, Git, GCC, GDB, and Linux kernel development, including targeted compatibility fixes across 13 source files and validation on real RTL8812BU hardware. Additional machine-learning and NLP experience supports a focused transition into AI security, adversarial research, and LLM security testing.

EXPERIENCE

Cybersecurity Researcher — Independent

Independent Cybersecurity Research & Security Engineering | Self-employed

Jul'24 – Present

- Conduct hands-on cybersecurity research across offensive and defensive security, threat intelligence, network security, web application security, wireless security, and system security.
- Perform reconnaissance, service enumeration, vulnerability assessment, privilege escalation, and controlled security testing using Nmap and Linux-based security tooling.
- Research adversary tradecraft, attack techniques, and TTPs; map findings to MITRE ATT&CK and analyze offensive techniques from a defensive perspective.
- Conduct web application security testing with Burp Suite and OWASP ZAP, including SQL injection, XSS, authentication, and access-control testing.
- Develop Python and Bash tooling for reconnaissance, security testing, proof-of-concept development, and repeatable research workflows.
- Analyze network traffic, packet captures, IOCs, suspicious communication patterns, and attacker behavior using Wireshark and related security tools.
- Conduct OSINT and security investigations involving domains, IP addresses, infrastructure, threat reports, malware, and Windows forensic artifacts.
- Practice security monitoring workflows including log analysis, alert triage, IOC identification, threat detection, and incident-response investigation in structured security labs.
- Maintain technical documentation covering security findings, testing methodology, attack paths, indicators, and remediation considerations.

Security Researcher — Independent

Independent Security Research & Security Engineering | Self-employed

Jul'26 – Aug'26

- Completed Linux kernel compatibility and wireless-security engineering work involving a Realtek RTL8812BU/RTL8822BU USB Wi-Fi driver.
- Analyzed driver source and kernel compatibility issues and implemented targeted fixes across 13 source files for newer Linux kernel APIs and build requirements.
- Diagnosed compiler, kernel API, and module-build issues using C, Linux kernel headers, GCC, conditional compilation, Git, and Linux kernel build workflows.
- Built and validated the modified kernel module on Kali Linux using real RTL8812BU hardware, confirming monitor mode, packet injection, packet capture, Wireshark, and Aircrack-ng workflows.

PROJECTS

Linux Kernel Compatibility & Wireless Security Engineering

C | Linux Kernel | RTL8812BU/RTL8822BU | Kali Linux | Git

Jul'26 – Aug'26

- Modified an existing Realtek USB Wi-Fi driver to address compatibility issues introduced by newer Linux kernel APIs and build requirements.
- Analyzed kernel-facing C code and implemented targeted compatibility changes across 13 source files while preserving the existing driver architecture and functionality.
- Diagnosed kernel-module build and API compatibility issues using C, Linux kernel headers, GCC, conditional compilation, Git, and Linux kernel build workflows.
- Built, installed, and validated the resulting kernel module on Kali Linux using real RTL8812BU hardware.
- Validated monitor mode, packet injection, packet capture, Wireshark analysis, and Aircrack-ng workflows as part of wireless-security testing.

Web Application Security Assessment Lab

PortSwigger Web Security Academy | Burp Suite | OWASP ZAP

Feb'26 – Present

- Conduct hands-on web application security testing covering SQL injection, cross-site scripting, authentication, access-control, and related OWASP Top 10 vulnerabilities.
- Analyze requests, responses, parameters, sessions, authentication mechanisms, and application behavior to identify vulnerability conditions and attack paths.
- Reproduce findings in controlled labs and document exploitation conditions, security impact, and remediation approaches.

- Perform reconnaissance, service enumeration, vulnerability assessment, system enumeration, and controlled exploitation across isolated virtual machines.
- Practice Linux privilege escalation and post-exploitation techniques and analyze attack paths from initial reconnaissance through elevated privileges.
- Validate vulnerabilities with controlled exploitation and document findings, exploitation paths, impact, and hardening considerations.

PROJECTS & TECHNICAL RESEARCH

Buffer Overflow Exploit Development

C / Linux / GDB / Exploit Development

Completed

- Analyzed stack-based buffer-overflow vulnerabilities and vulnerable program behavior in controlled Linux environments.
- Used GDB and memory-analysis techniques to examine stack layout, control flow, and exploitation conditions and developed proof-of-concept exploitation workflows.

Enterprise Network Penetration Testing Lab

Nmap / Metasploit / Linux / Network Security

Completed

- Conducted reconnaissance, service enumeration, vulnerability assessment, and attack-path analysis against a controlled enterprise-style network.
- Investigated exposed services and security weaknesses, validated findings through controlled testing, and documented attack paths and remediation considerations.
- Used Nmap and Metasploit to investigate vulnerable services and understand exploitation paths within an isolated lab environment.

Fake News Detection System — Capstone Project

Machine Learning / Python / Natural Language Processing

Apr'24

- Designed and implemented a machine-learning system to classify news articles as legitimate or fake using labeled datasets.
- Performed text preprocessing, feature extraction, model training, and evaluation to assess classification performance.

SKILLS

Offensive Security	Reconnaissance, Service Enumeration, Vulnerability Assessment, Penetration Testing, Exploit Development, Security Testing, Privilege Escalation, Post-Exploitation, Attack-Path Analysis
Threat & Adversary Research	Threat Intelligence, OSINT, TTP Analysis, MITRE ATT&CK, Adversary Research, Threat Report Analysis, Infrastructure Analysis
Web & API Security	OWASP Top 10, SQL Injection, Cross-Site Scripting, Authentication, Access Control, Burp Suite, OWASP ZAP, HTTP/HTTPS
Network & Wireless	TCP/IP, DNS, DHCP, Routing, Switching, Subnetting, CIDR, Network Traffic Analysis, Packet Capture, Monitor Mode, Packet Injection, Wireshark, Aircrack-ng
Forensics & Malware	Windows Forensics, Registry Analysis, File-System Analysis, Disk Analysis, Malware Analysis, Static/Dynamic Analysis
Programming Systems	& Python, Bash, C, SQL, Linux, Windows, Git, GCC, GDB, Linux Kernel Headers, Kernel Build Workflows
Security Tooling	Nmap, Burp Suite, OWASP ZAP, Wireshark, Metasploit, Nessus, GDB, Aircrack-ng
AI / ML Frameworks	& Machine Learning, NLP, Text Preprocessing, Feature Extraction, Classification, NIST Cybersecurity Framework, OWASP Top 10, MITRE ATT&CK

TOOLS & TECHNICAL STACK

Offensive Security Analysis	Nmap, Burp Suite, OWASP ZAP, Metasploit, Aircrack-ng, Nessus Wireshark, packet capture, network traffic analysis, IOC analysis, Windows forensic analysis
Programming Systems	Python, Bash, C, SQL Linux, Kali Linux, Windows, GCC, Linux kernel build workflows, Git
Security Frameworks	MITRE ATT&CK, OWASP Top 10, NIST Cybersecurity Framework

CORE SECURITY CAPABILITIES

Security Operations	Security Monitoring, Log Analysis, Alert Triage, Threat Detection, IOC Analysis, Incident Response Workflows, MITRE ATT&CK
Threat Intelligence	OSINT, Threat Intelligence, Threat Report Analysis, TTP Analysis, Infrastructure Analysis, Adversary Research
Vulnerability Assessment	Reconnaissance, Service Enumeration, Vulnerability Assessment, Security Testing, Privilege Escalation, Attack-Path Analysis
Web Security	OWASP Top 10, SQL Injection, Cross-Site Scripting, Authentication, Access Control, Burp Suite, OWASP ZAP
Network Security	TCP/IP, DNS, DHCP, HTTP/HTTPS, Routing, Switching, Subnetting, CIDR, Packet Capture, Traffic Analysis
Forensics / Malware	Windows Forensics, Registry Analysis, File-System Analysis, Disk Analysis, Static/Dynamic Malware Analysis
Systems / Programming Security Tools	Linux, Windows, Python, Bash, C, SQL, Git, GCC, Linux Kernel Build Workflows Nmap, Wireshark, Burp Suite, OWASP ZAP, Metasploit, Nessus, Aircrack-ng

CERTIFICATIONS & PROFESSIONAL TRAINING

Tata — Cybersecurity Analyst Job Simulation	Aug 2026 Forage
IAM strategy assessment, solution design, implementation planning, and secure access concepts. Google Cybersecurity Professional Certificate	Feb 2026 Coursera
Security, networking, Linux, SQL, Python, SIEM, threat detection, incident response, and SOC workflows. Key Learning: Log analysis, alert triage, vulnerability management, risk assessment, access control, and NIST Cybersecurity Framework. Introduction to Blockchain and Web3	May 2024 edX
Blockchain, decentralized systems, Web3 architecture, and security considerations. Natural Language Processing Specialization	Apr 2023 Coursera
Text preprocessing, classification, and sequence models using Python. Java Full Stack Developer Training	Feb 2023 CipherSchools
Java-based full-stack development and databases. Google Cloud Facilitator Program	May 2022 Google
Cloud computing and infrastructure.	

SELECTED TECHNICAL EVIDENCE

- **Linux kernel engineering:** analyzed kernel-facing C code, diagnosed API/build compatibility issues, implemented targeted fixes across 13 source files, and validated the resulting module on real RTL8812BU hardware.
- **Wireless security validation:** validated monitor mode, packet injection, packet capture, Wireshark traffic analysis, and Aircrack-ng workflows using physical RTL8812BU hardware on Kali Linux.
- **Web security testing:** used Burp Suite and OWASP ZAP to analyze HTTP requests/responses and test SQL injection, XSS, authentication, and access-control vulnerabilities in controlled environments.
- **Vulnerability assessment:** performed reconnaissance, service enumeration, vulnerability validation, Linux privilege escalation, post-exploitation analysis, and attack-path documentation across isolated virtual machines.
- **Exploit research:** used C, Linux, GDB, stack/memory analysis, and controlled proof-of-concept workflows to study buffer-overflow exploitation conditions.
- **Threat research and monitoring:** conducted OSINT and threat-report analysis, mapped TTPs to MITRE ATT&CK, and practiced log analysis, alert triage, IOC identification, threat detection, and incident-response investigation workflows.
- **Security tooling:** developed Python/Bash scripts for reconnaissance, repeatable testing, proof-of-concept development, and basic automation.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering	2024
Relevant Coursework: Network Security, Ethical Hacking, Penetration Testing, Machine Learning, Automation	Jalandhar, Punjab
Intermediate (MPC), Telangana Board of Intermediate Education	2020
Mathematics, Physics, Chemistry	Warangal, Telangana

ADDITIONAL TRAINING

- **Java Full Stack Developer Training** — Feb 2023 | CipherSchools — Java-based full-stack development and databases.
- **Google Cloud Facilitator Program** — May 2022 | Google — cloud computing and infrastructure.

TECHNICAL FOCUS

Security Operations	Security monitoring, log analysis, alert triage, IOC analysis, threat detection, incident-response investigation workflows
Offensive Security	Reconnaissance, enumeration, web application security testing, vulnerability validation, controlled exploitation, privilege escalation
Security Research	OSINT, threat intelligence, TTP analysis, MITRE ATT&CK mapping, vulnerability research, malware-analysis exercises
Systems Security	Linux, Windows, C, Python, Bash, kernel/module build workflows, wireless-security testing, network traffic analysis

RESEARCH AREAS

Offensive Research	Vulnerability discovery, exploit development, privilege escalation, post-exploitation, attack-path analysis, web/network testing
Systems Security	Linux internals, kernel compatibility, C systems programming, wireless security, packet analysis, driver engineering
Detection & Threat Research	Threat intelligence, OSINT, adversary TTPs, IOC analysis, log analysis, alert triage, MITRE ATT&CK
AI Security Direction	AI/ML security research, NLP, adversarial testing, prompt-injection/RAG security research track, red-team methodology

TARGET ROLES

Cybersecurity Researcher | Offensive Security | Security Engineer | SOC / Security Analyst | Vulnerability Assessment / VAPT | AI Security